

XUANREN SONG

Researcher - AI and Environmental Data Systems

Melbourne, Victoria, Australia · kowdataanalyze@gmail.com

LINKS

[Google Scholar](#) · [GitHub](#) · [Planetary Monitor](#)

PROFILE

Researcher and data engineer working across environmental data infrastructure, artificial intelligence, and multimodal systems. Experienced in automated global emissions and power-sector data products, geospatial and time-series modelling, and conversational access to Earth-system data. Current doctoral interests include representation learning for 3D human motion and embodied multimodal AI.

EMPLOYMENT HISTORY

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| Mar 2022 - Feb 2024 | <p>Research Assistant, Carbon Monitor Project, Tsinghua University</p> <ul style="list-style-type: none">• Developed automated Python pipelines and research data products for near-real-time carbon and power-sector monitoring.• Contributed to peer-reviewed studies and public-facing databases, including CarbonMonitor-Power as one of three equal contributors. |
| Jan 2016 - Jan 2019 | <p>Senior Billing Specialist, Chang Tsi & Partners</p> <ul style="list-style-type: none">• Coordinated international client billing requests and supporting document workflows.• Improved reporting and verification processes, reducing turnaround time from days to hours. |
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SELECTED PROJECTS

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| 2026 - Present | <p>Planetary Monitor and Planetary AI</p> <ul style="list-style-type: none">• Contributing to an integrated explorer for atmosphere, land, ocean, climate, and human-activity datasets across globe, map, and timeline views.• Researching and developing Planetary AI, a conversational interface for querying data and producing interpretable comparisons, charts, and maps. |
| 2024 - Present | <p>Carbon Monitor Data Infrastructure and CarbonMonitor-Power Website</p> <ul style="list-style-type: none">• Maintain automated infrastructure for the Carbon Monitor main platform, CarbonMonitor-Power, GRACED, Carbon Monitor Cities, and Carbon Monitor China.• Co-developed the CarbonMonitor-Power research dataset and independently developed its current 69-country website with automated daily generation and deployment. |
| Master's thesis | <p>X-TransformerCO2</p> <ul style="list-style-type: none">• Developed an explainable AI framework for daily national CO2-emissions forecasting, integrating energy, mobility, calendar, and event-related drivers. |

EDUCATION

2024 - 2025	Master of Artificial Intelligence, Monash University <i>Awarded with Distinction; thesis mark 90 (HD)</i>
2019 - 2020	Master of Business Analytics, Deakin University
2012 - 2016	Bachelor of Science, major in Economics, University of Oregon

TECHNICAL SKILLS	Programming and data systems: Python, SQL, FastAPI, Streamlit, Xarray, NetCDF, Linux, automated pipelines, APIs, and geospatial workflows. AI and research methods: machine learning, deep learning, time-series forecasting, explainable AI, multimodal systems, and reproducible research.
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SELECTED PEER-REVIEWED PUBLICATIONS

1. Biqing Zhu, Zhu Deng, **Xuanren Song**, Wenli Zhao, Da Huo, Taochun Sun, Piyu Ke, Duo Cui, Chenxi Lu, et al. (2023). *CarbonMonitor-Power near-real-time monitoring of global power generation on hourly to daily scales*. **Scientific Data**. [DOI](#) *Equal contribution*.
2. Shujie Zhang, Wenli Zhao, Biqing Zhu, Chunhua Yan, **Xuanren Song**, et al. (2025). *A near-real time daily European Power Consumption and Carbon Intensity Dataset (ECON-PowerCI)*. **Scientific Data**. [DOI](#)
3. Duo Cui, Siyao Yang, **Xuanren Song**, Xiaoting Huang, et al. (2025). *Monthly methane emissions in Chinese mainland provinces from 2013-2022*. **Scientific Data**. [DOI](#)
4. Chenxi Lu, Yingjian Huang, Ying Yu, Jiawei Hu, Huibin Mo, Yun Li, Da Huo, **Xuanren Song**, et al. (2024). *Health co-benefits of post-COVID-19 low-carbon recovery in Chinese cities*. **Nature Cities**. [DOI](#)
5. Lena Gurriaran, Yannig Goude, Katsumasa Tanaka, Biqing Zhu, Zhu Deng, **Xuanren Song**, and Philippe Ciais (2024). *Carbon Monitor Power-Simulators (CMP-SIM v1.0) across countries: a data-driven approach to simulate daily power generation*. **Geoscientific Model Development**. [DOI](#)
6. Xiaojuan Lin, Ronald van der A, Jos de Laat, Henk Eskes, Frederic Chevallier, Philippe Ciais, Zhu Deng, Yuanhao Geng, **Xuanren Song**, et al. (2023). *Monitoring and quantifying CO2 emissions of isolated power plants from space*. **Atmospheric Chemistry and Physics**. [DOI](#)
7. Piyu Ke, Zhu Deng, Biqing Zhu, Bo Zheng, Yilong Wang, Olivier Boucher, Simon Ben Arous, Chuanlong Zhou, Robbie M. Andrew, Xinyu Dou, Taochun Sun, **Xuanren Song**, et al. (2023). *Carbon Monitor Europe near-real-time daily CO2 emissions for 27 EU countries and the United Kingdom*. **Scientific Data**. [DOI](#)
8. Xinyu Dou, Jinpyo Hong, Philippe Ciais, Frederic Chevallier, Feifan Yan, Ying Yu, Yifan Hu, Da Huo, Yun Sun, Yilong Wang, Steven J. Davis, Monica Crippa, Greet Janssens-Maenhout, Diego Guizzardi, Efisio Solazzo, Xiaojuan Lin, **Xuanren Song**, et al. (2023). *Near-real-time global gridded daily CO2 emissions 2021*. **Scientific Data**. [DOI](#)
9. Chenxi Lu, W. Neil Adger, Karyn Morrissey, Shaohui Zhang, Sergey Venevsky, Hao Yin, Taochun Sun, **Xuanren Song**, et al. (2022). *Scenarios of demographic distributional aspects of health co-benefits from decarbonising urban transport*. **The Lancet Planetary Health**. [DOI](#)